NETWORKING COMMANDS:

Tcpdump

tcpdump is a powerful and widely used command-line packet sniffer or packet analyzer tool that is used to capture or filter TCP/IP packets that are received or transferred over a network on a specific interface for analysis.

```
susel:~ # tcpdump -i eth0
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 96 bytes
20:39:28.014065 IP 192.168.198.1.netbios-ns > 192.168.198.255.netbios-ns: NBT U
DP PACKET(137): QUERY; REQUEST; BROADCAST
20:39:28.014840 IP 192.168.198.128.56851 > 192.168.198.2.domain: 18867+ PTR? 25
5.198.168.192.in-addr.arpa. (46)
20:39:28.027418 IP 192.168.198.1.49733 > 224.0.0.252.llmnr: UDP, length 22
20:39:28.027850 IP 192.168.198.128.50611 > lhr14s24-in-f19.1e100.net.https: P 2
912329209:2912329246(37) ack 1375935787 win 18760
20:39:28.034322 IP lhr14s24-in-f19.1e100.net.https > 192.168.198.128.50611: . a
ck 37 win 64240
20:39:28.037196 IP6 fe80::2cfe:5154:6c0d:fafd.65460 > ff02::1:3.llmnr: UDP, len
gth 22
20:39:28.039057 IP 192.168.198.1.65460 > 224.0.0.252.llmnr: UDP, length 22
20:39:28.051576 IP 192.168.198.2.domain > 192.168.198.128.56851: 18867 NXDomain
 0/1/0 (95)
20:39:28.051744 IP 192.168.198.128.35496 > 192.168.198.2.domain: 58919+ PTR? 1.
198.168.192.in-addr.arpa. (44)
```

Netstat

Displays active TCP connections, ports on which the computer is listening, Ethernet statistics, IP routing table, IPv4 statistics, and IPv6 statistics. It indicates the state of a TCP connection. it's a helpful tool in finding problems and determining the amount of traffic on the network as a performance measurement.

```
Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49674        SURYA:49675            ESTABLISHED
  TCP    127.0.0.1:49675        SURYA:49674            ESTABLISHED
  TCP    127.0.0.1:49676        SURYA:49677            ESTABLISHED
  TCP    127.0.0.1:49677        SURYA:49676            ESTABLISHED
  TCP    127.0.0.1:49689        SURYA:49690            ESTABLISHED
  TCP    127.0.0.1:49690        SURYA:49689            ESTABLISHED
  TCP    127.0.0.1:49692        SURYA:49693            ESTABLISHED
  TCP    127.0.0.1:49693        SURYA:49692            ESTABLISHED
  TCP    127.0.0.1:63192        SURYA:65001            ESTABLISHED
  TCP    127.0.0.1:63193        SURYA:63904            ESTABLISHED
  TCP    127.0.0.1:63904        SURYA:63193            ESTABLISHED
  TCP    127.0.0.1:65001        SURYA:63192            ESTABLISHED
```

ifconfig / ipconfig

Displays basic current TCP/IP network configuration. It is very useful to troubleshoot networking problems. ipconfig/all is used to provide detailed information such as IP address, subnet mask, MAC address, DNS server, DHCP server, default gateway, etc. ipconfig/renew is used to renew a DHCP-assigned IP address, whereas ipconfig/release is used to discard the assigned DHCP IP address.

```
Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b185:68b0:72aa:e9f0%11
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::f2c6:3071:485e:bb6a%20
   IPv4 Address. . . . . . . . . . . : 192.168.91.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

nslookup

It provides a command-line utility for querying DNS table of a DNS Server. It returns IP address for the given host name.

```
Default Server:  UnKnown
Address:  fec0:0:0:ffff::1
```

traceroute / tracert

Displays the path taken to a destination by sending ICMP Echo Request messages to the destination with TTL field values. The path displayed is the list of nearest router interfaces taken along each hop in the path between source host and destination

```
Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
               [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d                 Do not resolve addresses to hostnames.
    -h maximum_hops    Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                 Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                 Force using IPv4.
    -6                 Force using IPv6.
```

ping

Verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) Echo Request messages. The receipt of corresponding Echo Reply messages are displayed, along with round-trip times. Ping is the primary TCP/IP command used to troubleshoot connectivity, reachability, and name resolution.

```
Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
```